DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology
alansynn@gatech.edu · alansynn.com

Research Interests

My research builds interactive systems for authoring and controlling motion at the boundary of computer graphics and HCI. My CHI 2026 system MotionSmith connects motion sketching, generated mechanisms, and fabrication-ready export, developed through a year-long participatory design process with expert automata makers. My current question: how can a creator make local motion edits while preserving accepted timing, contacts, and unaffected behavior?

Education

Georgia Tech, Ph.D. in Computer Science — Aug. 2022-Present
- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE — Mar. 2020-Feb. 2022
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE — Mar. 2012-Feb. 2020

Experience

Ph.D. Research Intern, Dolby Laboratories — May 2024-Aug. 2025
- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. — June-Aug. 2023
- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. — Jan. 2021-Aug. 2022
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador — 2020-2022
- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc. — Jan.-Dec. 2019
- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio) — Apr. 2017-Jan. 2019
- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

1 **DoangJoo Synn**, Zhifan Guo, Sehoon Ha, and HyunJoo Oh. "MotionSmith: A Sketch-Based Design System for Automata Making." *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi:10.1145/3772318.3791545.

2 Jin Yu, Poojita Garg, **DoangJoo Synn**, and HyunJoo Oh. "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning." *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi:10.1145/3706598.3713260.